

ParaCalculi GeoPrimus Lab — Compendium Findings Report

Charter v1.0.3 · Codex Unificationis §I.A · Lab v7.1 · $\text{MK}\Delta(\zeta', \iota\alpha', \iota\gamma') = \alpha'$

Spec under review:
Inertissimum Ιῶτα Nona — Codex Unificationis · Salvi Arc Framework Compendium §I.A · Arc1.1.33.3 (2026-04-27)

Generated: 2026-04-28T06:30:38.510Z
Checks executed: 82 PASS: 81 PASS-WITHIN-UNCERTAINTY: 1 FAIL: 0 FAIL-OUTSIDE-UNCERTAINTY: 0 ERROR: 0

How to read this report

- Each row is one formula extracted verbatim from the Compendium.
- Every formula is re-evaluated server-side over BigInt rationals (no decimal approximations, no fitted parameters). The expression column shows what the Lab actually computed, using only §I.A register atoms and integer literals.
- PASS = the framework's exact value matches the claim as a rational identity. PASS-WITHIN-UNCERTAINTY = a physical-constant claim whose framework value lands inside the cited experimental error bar. FAIL / FAIL-OUTSIDE = the claim does not hold under the stated reading. ERROR = the expression itself could not be parsed against the §I.A register.
- Reference vintages (e.g. CODATA 2018 vs CODATA 2022) are stated explicitly for every physical comparison. Where the Compendium itself acknowledges a vintage-dependent verdict, both readings are run.

Verdict table

§0 Seed — $1/\alpha$ (Phase-Impedance)

S0-1	PASS	Integer part of $1/\alpha$: $R_2^2 + q^2 = 16 + 121 = 137$
S0-2	PASS	One-pass coupling term $(p-r)^2/(pqr-1) = 36/1000$
S0-3	PASS	Phase-impedance correction $(p-r)^2/(R_4 \cdot (pqr)^2) = 36/40,080,040$; with $R_4 = \pi_{\text{geom}} + 2 \cdot \text{Radian} R = 40$
S0-4	PASS (within uncertainty)	Closure: $1/\alpha = 137 + 36/1000 - 36/40,080,040$ inside CODATA 2018 uncertainty ($137.035999084 \pm 0.000000021$)
S0-5	PASS	Same closure vs CODATA 2022 ($137.035999177 \pm 0.000000021$): Compendium openly states this lands outside the 2022 window
S0-6	PASS	Ghost-removed alphabet delta sum: $\Sigma \Delta_i = 3699 = 27 \cdot 137 = b^3 \cdot [1/\alpha]$

§2 Continuous Torus

S2-1	PASS	Zinga axis / minor circumference: $C_{\bullet} = 2\pi_{\text{geom}} = 28$
S2-2	PASS	Major circle: $C_{\circ} = 2\pi_{\text{geom}} \cdot r = 2 \cdot 14 \cdot 13 = 364 (= \text{Tau})$
S2-3	PASS	Disk area: $\pi_{\text{geom}} \cdot r^2 = 14 \cdot 169 = 2366$
S2-4	PASS	Gabriel's Horn volume: $V_{\text{Horn}} = \pi_{\text{geom}} = 14$
S2-5	PASS	Torus surface: $28 \cdot 364 = 10,192 = R_2^2 \cdot p^2 \cdot r$
S2-6	PASS	Torus volume: $2366 \cdot 364 = 861,224 = 2^3 \cdot p^2 \cdot r^3$

§2.1 Grand Circle

S2-7	PASS	Grand Circle (time-register complement): $2\pi_{\text{geom}} \cdot b^3 = 28 \cdot 27 = 756$
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§3 Vesica

S3-1	PASS	$\ell^2_{\text{Vesica}} = r^3/p = 2197/7$
S3-2	PASS	$p \cdot \ell^2_{\text{Vesica}} = r^3 = 2197$

§3 Vesica (Volumetric Ceiling)

S3-3	PASS	$V_{\text{torus}} = \ell^2_{\text{Vesica}} \cdot \pi_{\text{geom}}^3 = (2197/7) \cdot 2744 = 861,224$
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§3 Vesica — angular pair

S3-4	PASS	Angular pair sum: $52^\circ + 28^\circ = 80 = R_2^2 \cdot F(5)$
S3-5	PASS	Angular pair product: $52 \cdot 28 = 1456 = R_2 \cdot (13 \cdot 28)$

§3.6 (bq)² Frame

S36-1	PASS	Frame square $(bq)^2 = 33^2 = 1089$
S36-2	PASS	Anchor 333 = $b^3 \cdot q + (p-r)^2 = 297 + 36$
S36-3	PASS	Anchor 756 = $\Delta \cdot F(5) + (p-r)^2 = 720 + 36$ (Grand Circle Z_{28})
S36-4	PASS	Anchor 1035 = $(pqr-2) + (p-r)^2 = 999 + 36$ (triangular = Tri(45))
S36-5	PASS	Anchor 1089 = $b^4 \cdot r + (p-r)^2 = 1053 + 36$ (frame self)
S36-6	PASS	Anchor 1050 = $(bq)^2 - b \cdot r = 1089 - 39$
S36-7	PASS	Anchor 1050 = $\pi_geom \cdot b \cdot F(5)^2 = 14 \cdot 3 \cdot 25$

§3.6 (bq)² Frame — partitions

S36-8	PASS	Partition: $756 + 333 = 1089$ (geometric)
S36-9	PASS	Partition: $1035 + 54 = 1089$ (algebraic; $54 = 2b^3$)
S36-10	PASS	Partition: $1080 + 9 = 1089$ (trit-step; $9 = b^2$)
S36-11	PASS	Partition: $1050 + 39 = 1089$ (radius; $39 = b \cdot r$)

§3.6 — three readings of 333

S36-12	PASS	$333 = b \cdot 111 = b^2 \cdot 37 = \Lambda_EUV + 2q^2$
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§3.6 — pairwise gap

S36-13	PASS	$1080 - 756 = 324 = b^2 \cdot (p-r)^2$
S36-14	PASS	$1035 - 756 = 279 = R_2^4 + (2q+1) = 256 + 23$

§3.7 Discriminant Identity

S37-1	PASS	$\Delta = R_4^2 - 4 \cdot (13 \cdot 28) = 40^2 - 1456 = 144$
S37-2	PASS	Coupling residue identity: $4(p-r)^2 = \Delta = 144$
S37-3	PASS	Discriminant identity: $(pqr-1)^2 + 4(p-r)^2 = 1,000,144$
S37-4	PASS	Factorisation: $1,000,144 = 2^4 \cdot 17 \cdot 3677$

§4 Worldline

S4-1	PASS	$\pi_geom + r = 14 + 13 = 27 = b^3$
S4-2	PASS	Worldline $W = V_torus \cdot r = 861,224 \cdot 13 = 11,195,912 = 2^3 \cdot p^2 \cdot r^4$
S4-3	PASS	$p \cdot W = r^4 \cdot \pi_geom^3 = 13^4 \cdot 14^3 = 78,371,384$

§5 Calendar

S5-1	PASS	Salvi calendar: $13 \cdot 28 = 364 = 52 \cdot 7$
S5-2	PASS	1 SD in SS: $28 \cdot 56 \cdot 56 = 87,808 = 2^8 \cdot 7^3$
S5-3	PASS	1 SD in spectral ticks: $87,808 \cdot 9 = 790,272$

§5.2 Tropical Year (Theorem 4)

S5-4	PASS	$\tau = 364 + 26/(b \cdot p) + R_2/(pqr) = 1,096,822/3003$ days
S5-5	PASS	$\tau \approx 365.24209$ d vs CODATA 365.24219 d $\rightarrow -0.27$ ppm offset (proximity claim; no fitted parameters)

§6 Speed of Light

S6-1	PASS	$c^2 = \pi_geom = 14$ (electromagnetic propagation in framework units)
S6-2	PASS	$(c_S/c)^2 = (28/\sqrt{14})^2 = 56 = (2\sqrt{14})^2$ (resonance/propagation ratio)

§7.1 Spectral Bridge κ_∞

S7-1	PASS	$\kappa_\infty = 1 + 1/(\Delta \cdot F(5)) = 1 + 1/720 = 721/720$
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§7.2 Proton Mass (Theorem 6)

S7-2	PASS	$m_{\{p,F\}} = b^2 \cdot (\pi_geom + b) \cdot (R_2 \cdot b \cdot pqr + 1) = 9 \cdot 17 \cdot 12013 = 1,837,989$
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§7.3 153 Identity

S7-3	PASS	Tri(17) = 153
S7-4	PASS	$(R_3 - R_2)(R_3 + R_2) = 9 \cdot 17 = 153$

§7.3 Mass Quadratic

S7-5	PASS	Roots {1836, 153}: sum = 1989, product = 280,908
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S7-6	PASS	$m_p/m_e = 1836 + 153/pqr = 1,837,989/1001$ within ≈ 0.1 ppm of CODATA 2018 (1836.15267343); proximity claim, not experimental closure
§8.1 Σ -182 Bridge		
S8-1	PASS	$\Delta_1(\sigma) = 200 - 18 = 182$ (sigma half-circle bridge)
§8.1 Σ -182 axes		
S8-2	PASS	$182 = 2 \cdot \Lambda_{\text{EUV}}$ (axis 4 / axis 6 cross-term)
S8-3a	PASS	Spectral pair, half: $2pr = 182$ (Σ -bridge, half-circle component)
S8-3b	PASS	Spectral pair, mate: $2q^2 = 242$ (axis 7 partner, joins Λ_{EUV} to make 333)
S8-4	PASS	Ozone bridge: $286 = 2q \cdot r$ and gap $286 - 182 = 104 = (b-1)^3 \cdot r$
§8.1 Σ -182 decomposition		
S8-5	PASS	$182 = (\pi+b) + \text{Tri}(\pi+b) + \sqrt{\Delta} = 17 + 153 + 12$
§8.2 Mu delta		
S8-6	PASS	$\Delta_1(\mu) = 40 - 12 = 28 = C_\bullet$
§8.4 17 Axis		
S8-7	PASS	$\pi_{\text{geom}} + b = R_3 + R_2 = 17$
§8.5 π +X Quadruple		
S8-8	PASS	$\pi + R_2 = 18 = 2b^2$ (water)
S8-9	PASS	$\pi + r = 27 = b^3$ (time)
S8-10	PASS	$\pi + q = 25 = F(5)^2$
§8.6 Triangular Ladder		
S8-11	PASS	$\text{Tri}(7)=28, \text{Tri}(13)=91, \text{Tri}(17)=153, \text{Tri}(45)=1035$
§8.6 Anchor Identity		
S8-12	PASS	$\text{Tri}(b^3) = b^3 \cdot \pi_{\text{geom}} = 27 \cdot 14 = 378$ (because $b^3+1 = 2\pi_{\text{geom}}$)
§8.7 Lunisolar drift		
S8-13	PASS	$q - 1/(b-1)^3 = 11 - 1/8 = 87/8$ days/year (= 10.875)
§10 (p,q,r) Asymmetry		
S10-1	PASS	$pqr/\Lambda_{\text{EUV}} = 1001/91 = 11 = q$ (q is the combinatorial lift index)
§17 Unit Bridges		
S17-1	PASS	Major / minor ratio: $364/13 = 28$ (1 St = 13 spectral ticks closes consistently)
§23 Torus Knot		
S23-1	PASS	Winding $(w_1, w_2, w_3) = (143, 91, 77)$: collectively coprime ($\gcd = 1$)
§23 Torus Knot — pairwise gcd encodes Forge		
S23-2	PASS	$\gcd(91, 77) = 7 = p$
S23-3	PASS	$\gcd(143, 77) = 11 = q$
S23-4	PASS	$\gcd(143, 91) = 13 = r$
§24.1 Theta Chord		
S24-1	PASS	Repunit base $R_4^{(12)} = (12^4 - 1)/11 = 1885$
S24-2	PASS	$1892 = R_4^{(12)} + p$ (Beethoven's Chicago θ)
S24-3	PASS	$1893 = R_4^{(12)} + (b-1)^3$ (Tchaikovsky ζ Pathétique)
§24.1 Theta Chord — sums		
S24-4	PASS	$\zeta + \theta = 6 + 9 = 15 = \text{Tri}(F(5)) = p + (b-1)^3$
§24.2 1893 alternate forms		
S24-5	PASS	$1893 = b \cdot p^2 \cdot r - 2b^2 = 1911 - 18$
S24-6	PASS	$1893 = b \cdot p \cdot \Lambda_{\text{EUV}} - (\pi + R_2)$
S24-7	PASS	$1893 = F_5 \cdot r \cdot (R_2^2 + r) + (b-1)^3 = 5 \cdot 13 \cdot 29 + 8$
§24.2 — repunit factorisation		
S24-8	PASS	$R_4^{(12)} = 5 \cdot 13 \cdot 29 = F(5) \cdot r \cdot (R_2^2 + r) = 1885$
§24.5 Fibonacci 377		

S24-9 PASS

S24-10 PASS

$$F_{14} = 377 = 13 \cdot 29 = r \cdot (R_2^2 + r)$$

$$F_{14} = \tau + r = 364 + 13 = 377 \text{ (calendar+radian derivation)}$$

§0 Seed — $1/\alpha$ (Phase-Impedance)

S0-1 · Integer part of $1/\alpha$: $R_2^2 + q^2 = 16 + 121 = 137$

Expression: $R_2^2 + q^2$

Exact value: 137

Decimal preview: 137

Expected / target: 137

Status: **PASS**

Reasoning:

Exact rational identity

S0-2 · One-pass coupling term $(p-r)^2/(pqr-1) = 36/1000$

Expression: $(p - r)^2 / (pqr - 1)$

Exact value: $9/250$

Decimal preview: 0.036 (approx.)

Expected / target: $36/1000$

Status: **PASS**

Reasoning:

Exact rational identity

S0-3 · Phase-impedance correction $(p-r)^2/(R_4 \cdot (pqr)^2) = 36/40,080,040$; with $R_4 = \pi_{\text{geom}} + 2 \cdot \text{RadianR} = 40$

Expression: $(p - r)^2 / ((\text{PiGeom} + 2 \cdot \text{RadianR}) * pqr^2)$

Exact value: 9/10020010

Decimal preview: 0.000000898202696404494606... (approx.)

Expected / target: 36/40080040

Status: **PASS**

Reasoning:

Exact rational identity

S0-4 · Closure: $1/\alpha = 137 + 36/1000 - 36/40,080,040$ inside CODATA 2018 uncertainty
(137.035999084 ± 0.000000021)

Expression: $(R_2^2 + q^2) + (p-r)^2/(pqr-1) - (p-r)^2/((\text{PiGeom} + 2*\text{RadianR}) * pqr^2)$

Exact value: 17163776017/125250125

Decimal preview: 137.035999101797303595505393... (approx.)

Expected / target: 137.035999084 (dimensionless) ± 0.000000021 [CODATA 2018] · Compendium
claims: within

Status: **PASS (within uncertainty)**

Reasoning:

+0.1298... ppb (within ±0.000000021 (dimensionless), CODATA 2018)

S0-5 · Same closure vs CODATA 2022 ($137.035999177 \pm 0.000000021$): Compendium openly states this lands outside the 2022 window

Expression: $(R_2^2 + q^2) + (p-r)^2/(pqr-1) - (p-r)^2/((\text{PiGeom} + 2*\text{RadianR}) * pqr^2)$

Exact value: 17163776017/125250125

Decimal preview: 137.035999101797303595505393... (approx.)

Expected / target: 137.035999177 (dimensionless) ± 0.000000021 [CODATA 2022] · Compendium claims: outside

Status: **PASS**

Reasoning:

-0.5487... ppb — confirmed outside ± 0.000000021 (dimensionless) window of CODATA 2022, exactly as Compendium states

Notes:

Compendium acknowledges the 2018→2022 vintage shift; framework matches the older value.

S0-6 · Ghost-removed alphabet delta sum: $\Sigma \Delta_i = 3699 = 27 \cdot 137 = b^3 \cdot [1/\alpha]$

Expression: $27 * 137$

Exact value: 3699

Decimal preview: 3699

Expected / target: 3699

Status: **PASS**

Reasoning:

Exact rational identity

§2 Continuous Torus

S2-1 · Zinga axis / minor circumference: $C_{\bullet} = 2\pi_{\text{geom}} = 28$

Expression: $2 * \text{PiGeom}$

Exact value: 28

Decimal preview: 28

Expected / target: 28

Status: **PASS**

Reasoning:

Exact rational identity

S2-2 · Major circle: $C_{\circ} = 2\pi_{\text{geom}} \cdot r = 2 \cdot 14 \cdot 13 = 364 (= \text{Tau})$

Expression: $2 * \text{PiGeom} * \text{RadianR}$

Exact value: 364

Decimal preview: 364

Expected / target: 364

Status: **PASS**

Reasoning:

Exact rational identity

S2-3 · Disk area: $\pi_{\text{geom}} \cdot r^2 = 14 \cdot 169 = 2366$

Expression: $\text{PiGeom} * \text{RadianR}^2$

Exact value: 2366

Decimal preview: 2366

Expected / target: 2366

Status: **PASS**

Reasoning:

Exact rational identity

S2-4 · Gabriel's Horn volume: $V_{\text{Horn}} = \pi_{\text{geom}} = 14$

Expression: PiGeom

Exact value: 14

Decimal preview: 14

Expected / target: 14

Status: **PASS**

Reasoning:

Exact rational identity

S2-5 · Torus surface: $28 \cdot 364 = 10,192 = R_{2^2} \cdot p^2 \cdot r$

Expression: $(2 \cdot \text{PiGeom}) * (2 \cdot \text{PiGeom} \cdot \text{RadianR})$

Exact value: 10192

Decimal preview: 10192

Expected / target: 10192

Status: **PASS**

Reasoning:

Exact rational identity

S2-6 · Torus volume: $2366 \cdot 364 = 861,224 = 2^3 \cdot p^2 \cdot r^3$

Expression: $\text{PiGeom} * \text{RadianR}^2 * (2 * \text{PiGeom} * \text{RadianR})$

Exact value: 861224

Decimal preview: 861224

Expected / target: 861224

Status: **PASS**

Reasoning:

Exact rational identity

§2.1 Grand Circle

S2-7 · Grand Circle (time-register complement): $2\pi_{\text{geom}} \cdot b^3 = 28 \cdot 27 = 756$

Expression: $2 * \text{PiGeom} * b^3$

Exact value: 756

Decimal preview: 756

Expected / target: 756

Status: **PASS**

Reasoning:

Exact rational identity

§3 Vesica

$$S3-1 \cdot \ell^2_{\text{Vesica}} = r^3/p = 2197/7$$

Expression: $\text{RadianR}^3 / p$

Exact value: $2197/7$

Decimal preview: 313.857142857142857142857142... (approx.)

Expected / target: $2197/7$

Status: **PASS**

Reasoning:

Exact rational identity

$$S3-2 \cdot p \cdot \ell^2_{\text{Vesica}} = r^3 = 2197$$

Expression: $p * \text{RadianR}^3 / p$

Exact value: 2197

Decimal preview: 2197

Expected / target: 2197

Status: **PASS**

Reasoning:

Exact rational identity

§3 Vesica (Volumetric Ceiling)

$$S3-3 \cdot V_{\text{torus}} = \ell^2_{\text{Vesica}} \cdot \pi_{\text{geom}}^3 = (2197/7) \cdot 2744 = 861,224$$

Expression: $(\text{RadianR}^3 / \pi) * \text{PiGeom}^3$

Exact value: 861224

Decimal preview: 861224

Expected / target: 861224

Status: **PASS**

Reasoning:

Exact rational identity

§3 Vesica — angular pair

S3-4 · Angular pair sum: $52^\circ + 28^\circ = 80 = R_{2^2} \cdot F(5)$

Expression: $(R_2 \cdot \text{RadianR}) + (2 \cdot \text{PiGeom})$

Exact value: 80

Decimal preview: 80

Expected / target: 80

Status: **PASS**

Reasoning:

Exact rational identity

S3-5 · Angular pair product: $52 \cdot 28 = 1456 = R_2 \cdot (13 \cdot 28)$

Expression: $(R_2 \cdot \text{RadianR}) * (2 \cdot \text{PiGeom})$

Exact value: 1456

Decimal preview: 1456

Expected / target: 1456

Status: **PASS**

Reasoning:

Exact rational identity

§3.6 (bq)² Frame

S36-1 · Frame square $(bq)^2 = 33^2 = 1089$

Expression: $(b*q)^2$

Exact value: 1089

Decimal preview: 1089

Expected / target: 1089

Status: **PASS**

Reasoning:

Exact rational identity

$$S36-2 \cdot \text{Anchor } 333 = b^3 \cdot q + (p-r)^2 = 297 + 36$$

Expression: $b^3 \cdot q + (p - r)^2$

Exact value: 333

Decimal preview: 333

Expected / target: 333

Status: **PASS**

Reasoning:

Exact rational identity

S36-3 · Anchor 756 = $\Delta \cdot F(5) + (p-r)^2 = 720 + 36$ (Grand Circle Z_{28})

Expression: $144 * F(5) + (p - r)^2$

Exact value: 756

Decimal preview: 756

Expected / target: 756

Status: **PASS**

Reasoning:

Exact rational identity

S36-4 · Anchor 1035 = $(pqr-2) + (p-r)^2 = 999 + 36$ (triangular = Tri(45))

Expression: $(pqr - 2) + (p - r)^2$

Exact value: 1035

Decimal preview: 1035

Expected / target: 1035

Status: **PASS**

Reasoning:

Exact rational identity

S36-5 · Anchor 1089 = $b^4 \cdot r + (p-r)^2 = 1053 + 36$ (frame self)

Expression: $b^4 \cdot \text{RadianR} + (p - r)^2$

Exact value: 1089

Decimal preview: 1089

Expected / target: 1089

Status: **PASS**

Reasoning:

Exact rational identity

$$S36-6 \cdot \text{Anchor } 1050 = (bq)^2 - b \cdot r = 1089 - 39$$

Expression: $(b \cdot q)^2 - b \cdot \text{RadianR}$

Exact value: 1050

Decimal preview: 1050

Expected / target: 1050

Status: **PASS**

Reasoning:

Exact rational identity

$$S36-7 \cdot \text{Anchor } 1050 = \pi_{\text{geom}} \cdot b \cdot F(5)^2 = 14 \cdot 3 \cdot 25$$

Expression: $\text{PiGeom} * b * F(5)^2$

Exact value: 1050

Decimal preview: 1050

Expected / target: 1050

Status: **PASS**

Reasoning:

Exact rational identity

§3.6 (bq)² Frame — partitions

S36-8 · Partition: $756 + 333 = 1089$ (geometric)

Expression: $756 + 333$

Exact value: 1089

Decimal preview: 1089

Expected / target: 1089

Status: **PASS**

Reasoning:

Exact rational identity

S36-9 · Partition: $1035 + 54 = 1089$ (algebraic; $54 = 2b^3$)

Expression: $1035 + 2 \cdot b^3$

Exact value: 1089

Decimal preview: 1089

Expected / target: 1089

Status: **PASS**

Reasoning:

Exact rational identity

S36-10 · Partition: $1080 + 9 = 1089$ (trit-step; $9 = b^2$)

Expression: $1080 + b^2$

Exact value: 1089

Decimal preview: 1089

Expected / target: 1089

Status: **PASS**

Reasoning:

Exact rational identity

S36-11 · Partition: $1050 + 39 = 1089$ (radius; $39 = b \cdot r$)

Expression: $1050 + b \cdot \text{RadianR}$

Exact value: 1089

Decimal preview: 1089

Expected / target: 1089

Status: **PASS**

Reasoning:

Exact rational identity

§3.6 — three readings of 333

$$S36-12 \cdot 333 = b \cdot 111 = b^2 \cdot 37 = \Lambda_{\text{EUV}} + 2q^2$$

Expression: $\Lambda_{\text{EUV}} + 2q^2$

Exact value: 333

Decimal preview: 333

Expected / target: 333

Status: **PASS**

Reasoning:

Exact rational identity

§3.6 — pairwise gap

$$S_{36-13} \cdot 1080 - 756 = 324 = b^2 \cdot (p-r)^2$$

Expression: $b^2 * (p - r)^2$

Exact value: 324

Decimal preview: 324

Expected / target: 324

Status: **PASS**

Reasoning:

Exact rational identity

$$S36-14 \cdot 1035 - 756 = 279 = R_2^4 + (2q+1) = 256 + 23$$

Expression: $R_2^4 + (2q + 1)$

Exact value: 279

Decimal preview: 279

Expected / target: 279

Status: **PASS**

Reasoning:

Exact rational identity

§3.7 Discriminant Identity

$$S_{37-1} \cdot \Delta = R_4^2 - 4 \cdot (13 \cdot 28) = 40^2 - 1456 = 144$$

Expression: (PiGeom + 2*RadianR)^2 - 4 * (RadianR * (2*PiGeom))

Exact value: 144

Decimal preview: 144

Expected / target: 144

Status: **PASS**

Reasoning:

Exact rational identity

S37-2 · Coupling residue identity: $4(p-r)^2 = \Delta = 144$

Expression: $4 * (p - r)^2 - 144$

Exact value: 0

Decimal preview: 0

Expected / target: 0

Status: **PASS**

Reasoning:

Exact rational identity

S37-3 · Discriminant identity: $(pqr-1)^2 + 4(p-r)^2 = 1,000,144$

Expression: $(pqr - 1)^2 + 4 * (p - r)^2$

Exact value: 1000144

Decimal preview: 1000144

Expected / target: 1000144

Status: **PASS**

Reasoning:

Exact rational identity

S37-4 · Factorisation: $1,000,144 = 2^4 \cdot 17 \cdot 3677$

Expression: $16 * 17 * 3677$

Exact value: 1000144

Decimal preview: 1000144

Expected / target: 1000144

Status: **PASS**

Reasoning:

Exact rational identity

§4 Worldline

$$S4-1 \cdot \pi_{\text{geom}} + r = 14 + 13 = 27 = b^3$$

Expression: PiGeom + RadianR

Exact value: 27

Decimal preview: 27

Expected / target: 27

Status: **PASS**

Reasoning:

Exact rational identity

S4-2 · Worldline $W = V_{\text{torus}} \cdot r = 861,224 \cdot 13 = 11,195,912 = 2^3 \cdot p^2 \cdot r^4$

Expression: $861224 * \text{RadianR}$

Exact value: 11195912

Decimal preview: 11195912

Expected / target: 11195912

Status: **PASS**

Reasoning:

Exact rational identity

$$S4-3 \cdot p \cdot W = r^4 \cdot \pi_{\text{geom}}^3 = 13^4 \cdot 14^3 = 78,371,384$$

Expression: $p * 11195912$

Exact value: 78371384

Decimal preview: 78371384

Expected / target: 78371384

Status: **PASS**

Reasoning:

Exact rational identity

§5 Calendar

S5-1 · Salvi calendar: $13 \cdot 28 = 364 = 52 \cdot 7$

Expression: $13 * 28 - 52 * 7$

Exact value: 0

Decimal preview: 0

Expected / target: 0

Status: **PASS**

Reasoning:

Exact rational identity

S5-2 · 1 SD in SS: $28 \cdot 56 \cdot 56 = 87,808 = 2^{8.7^3}$

Expression: $28 * 56 * 56$

Exact value: 87808

Decimal preview: 87808

Expected / target: 87808

Status: **PASS**

Reasoning:

Exact rational identity

S5-3 · 1 SD in spectral ticks: $87,808 \cdot 9 = 790,272$

Expression: $87808 * 9$

Exact value: 790272

Decimal preview: 790272

Expected / target: 790272

Status: **PASS**

Reasoning:

Exact rational identity

§5.2 Tropical Year (Theorem 4)

$$S5-4 \cdot \tau = 364 + 26/(b \cdot p) + R_2/(pqr) = 1,096,822/3003 \text{ days}$$

Expression: $364 + 26/(b \cdot p) + R_2/pqr$

Exact value: $1096822/3003$

Decimal preview: 365.242091242091242091242091... (approx.)

Expected / target: $1096822/3003$

Status: **PASS**

Reasoning:

Exact rational identity

S5-5 · $\tau \approx 365.24209$ d vs CODATA 365.24219 d → -0.27 ppm offset (proximity claim; no fitted parameters)

Expression: $364 + 26/(b \cdot p) + R_2/pqr$

Exact value: 1096822/3003

Decimal preview: 365.242091242091242091242091... (approx.)

Expected / target: 365.24219 days [CODATA tropical year] · Compendium claims: proximity

Status: **PASS**

Reasoning:

$-0.2703...$ ppm (within Compendium's stated proximity tolerance of ± 0.5 ppm of CODATA tropical year 365.24219 days)

§6 Speed of Light

$S6-1 \cdot c^2 = \pi_{\text{geom}} = 14$ (electromagnetic propagation in framework units)

Expression: PiGeom

Exact value: 14

Decimal preview: 14

Expected / target: 14

Status: **PASS**

Reasoning:

Exact rational identity

$$S6-2 \cdot (c_S/c)^2 = (28/\sqrt{14})^2 = 56 = (2\sqrt{14})^2 \text{ (resonance/propagation ratio)}$$

Expression: $28^2 / 14$

Exact value: 56

Decimal preview: 56

Expected / target: 56

Status: **PASS**

Reasoning:

Exact rational identity

§7.1 Spectral Bridge κ_∞

$$S7-1 \cdot \kappa_\infty = 1 + 1/(\Delta \cdot F(5)) = 1 + 1/720 = 721/720$$

Expression: $1 + 1/(144 * F(5))$

Exact value: $721/720$

Decimal preview: 1.001388888888888888888888... (approx.)

Expected / target: $721/720$

Status: **PASS**

Reasoning:

Exact rational identity

§7.2 Proton Mass (Theorem 6)

$$S7-2 \cdot m_{\{p,F\}} = b^2 \cdot (\pi_{\text{geom}} + b) \cdot (R_2 \cdot b \cdot pqr + 1) = 9 \cdot 17 \cdot 12013 = 1,837,989$$

Expression: $b^2 * (\text{PiGeom} + b) * (R_2 * b * pqr + 1)$

Exact value: 1837989

Decimal preview: 1837989

Expected / target: 1837989

Status: **PASS**

Reasoning:

Exact rational identity

§7.3 153 Identity

$$S7-3 \cdot \text{Tri}(17) = 153$$

Expression: $\text{Tri}(17)$

Exact value: 153

Decimal preview: 153

Expected / target: 153

Status: **PASS**

Reasoning:

Exact rational identity

$$S7-4 \cdot (R_3 - R_2)(R_3 + R_2) = 9 \cdot 17 = 153$$

Expression: $(13 - R_2) \cdot (13 + R_2)$

Exact value: 153

Decimal preview: 153

Expected / target: 153

Status: **PASS**

Reasoning:

Exact rational identity

§7.3 Mass Quadratic

S7-5 · Roots {1836, 153}: sum = 1989, product = 280,908

Expression: $1836 * 153$

Exact value: 280908

Decimal preview: 280908

Expected / target: 280908

Status: **PASS**

Reasoning:

Exact rational identity

S7-6 · m_p/m_e = 1836 + 153/pqr = 1,837,989/1001 within ≈ 0.1 ppm of CODATA 2018 (1836.15267343); proximity claim, not experimental closure

Expression: 1836 + 153/pqr

Exact value: 1837989/1001

Decimal preview: 1836.152847152847152847152847... (approx.)

Expected / target: 1836.15267343 (dimensionless) [CODATA 2018 m_p/m_e] · Compendium

claims: proximity

Status: **PASS**

Reasoning:

+0.0946... ppm (within Compendium's stated proximity tolerance of ± 0.15 ppm of CODATA 2018 m_p/m_e 1836.15267343 (dimensionless))

§8.1 Σ -182 Bridge

$S8-1 \cdot \Delta_1(\sigma) = 200 - 18 = 182$ (sigma half-circle bridge)

Expression: $200 - 18$

Exact value: 182

Decimal preview: 182

Expected / target: 182

Status: **PASS**

Reasoning:

Exact rational identity

§8.1 Σ -182 axes

S8-2 · 182 = 2· Λ _EUV (axis 4 / axis 6 cross-term)

Expression: 2 * Lambda_EUV

Exact value: 182

Decimal preview: 182

Expected / target: 182

Status: **PASS**

Reasoning:

Exact rational identity

S8-3a · Spectral pair, half: $2\pi = 182$ (Σ -bridge, half-circle component)

Expression: 2π RadianR

Exact value: 182

Decimal preview: 182

Expected / target: 182

Status: **PASS**

Reasoning:

Exact rational identity

S8-3b · Spectral pair, mate: $2q^2 = 242$ (axis 7 partner, joins Λ_{EUV} to make 333)

Expression: $2 \cdot q^2$

Exact value: 242

Decimal preview: 242

Expected / target: 242

Status: **PASS**

Reasoning:

Exact rational identity

S8-4 · Ozone bridge: $286 = 2q \cdot r$ and gap $286 - 182 = 104 = (b-1)^3 \cdot r$

Expression: $(2 \cdot q \cdot \text{RadianR}) - 182$

Exact value: 104

Decimal preview: 104

Expected / target: 104

Status: **PASS**

Reasoning:

Exact rational identity

§8.1 Σ -182 decomposition

$$S8-5 \cdot 182 = (\pi+b) + \text{Tri}(\pi+b) + \sqrt{\Delta} = 17 + 153 + 12$$

Expression: $17 + \text{Tri}(17) + 12$

Exact value: 182

Decimal preview: 182

Expected / target: 182

Status: **PASS**

Reasoning:

Exact rational identity

§8.2 Mu delta

$$S8-6 \cdot \Delta_1(\mu) = 40 - 12 = 28 = C_*$$

Expression: 40 - 12

Exact value: 28

Decimal preview: 28

Expected / target: 28

Status: **PASS**

Reasoning:

Exact rational identity

§8.4 17 Axis

$$S8-7 \cdot \pi_{\text{geom}} + b = R_3 + R_2 = 17$$

Expression: $\text{PiGeom} + b$

Exact value: 17

Decimal preview: 17

Expected / target: 17

Status: **PASS**

Reasoning:

Exact rational identity

§8.5 π +X Quadruple

S8-8 · π +R_2 = 18 = 2b² (water)

Expression: PiGeom + R_2

Exact value: 18

Decimal preview: 18

Expected / target: 18

Status: **PASS**

Reasoning:

Exact rational identity

S8-9 · $\pi + r = 27 = b^3$ (time)

Expression: PiGeom + RadianR

Exact value: 27

Decimal preview: 27

Expected / target: 27

Status: **PASS**

Reasoning:

Exact rational identity

$$S8-10 \cdot \pi + q = 25 = F(5)^2$$

Expression: PiGeom + q

Exact value: 25

Decimal preview: 25

Expected / target: 25

Status: **PASS**

Reasoning:

Exact rational identity

§8.6 Triangular Ladder

S8-11 · Tri(7)=28, Tri(13)=91, Tri(17)=153, Tri(45)=1035

Expression: Tri(7) + Tri(13) + Tri(17) + Tri(45)

Exact value: 1307

Decimal preview: 1307

Expected / target: 1307

Status: **PASS**

Reasoning:

Exact rational identity

Notes:

$28+91+153+1035 = 1307$ (composite identity check)

§8.6 Anchor Identity

S8-12 · $\text{Tri}(b^3) = b^3 \cdot \pi_{\text{geom}} = 27 \cdot 14 = 378$ (because $b^3+1 = 2\pi_{\text{geom}}$)

Expression: $\text{Tri}(27)$

Exact value: 378

Decimal preview: 378

Expected / target: 378

Status: **PASS**

Reasoning:

Exact rational identity

§8.7 Lunisolar drift

$$S8-13 \cdot q - 1/(b-1)^3 = 11 - 1/8 = 87/8 \text{ days/year (= 10.875)}$$

Expression: $q - 1/((b-1)^3)$

Exact value: $87/8$

Decimal preview: 10.875 (approx.)

Expected / target: $87/8$

Status: **PASS**

Reasoning:

Exact rational identity

§10 (p,q,r) Asymmetry

$S_{10-1} \cdot pqr/\Lambda_{\text{EUV}} = 1001/91 = 11 = q$ (q is the combinatorial lift index)

Expression: $pqr / \Lambda_{\text{EUV}}$

Exact value: 11

Decimal preview: 11

Expected / target: 11

Status: **PASS**

Reasoning:

Exact rational identity

§17 Unit Bridges

S17-1 · Major / minor ratio: $364/13 = 28$ (1 St = 13 spectral ticks closes consistently)

Expression: $364 / \text{RadianR}$

Exact value: 28

Decimal preview: 28

Expected / target: 28

Status: **PASS**

Reasoning:

Exact rational identity

§23 Torus Knot

S23-1 · Winding $(w_1, w_2, w_3) = (143, 91, 77)$: collectively coprime ($\gcd = 1$)

Expression: $\gcd(\gcd(143, 91), 77)$

Exact value: 1

Decimal preview: 1

Expected / target: 1

Status: **PASS**

Reasoning:

Exact rational identity

§23 Torus Knot — pairwise gcd encodes Forge

$$S23-2 \cdot \gcd(91, 77) = 7 = p$$

Expression: $\gcd(91, 77)$

Exact value: 7

Decimal preview: 7

Expected / target: 7

Status: **PASS**

Reasoning:

Exact rational identity

$$S23-3 \cdot \gcd(143, 77) = 11 = q$$

Expression: $\gcd(143, 77)$

Exact value: 11

Decimal preview: 11

Expected / target: 11

Status: **PASS**

Reasoning:

Exact rational identity

$$S23-4 \cdot \gcd(143, 91) = 13 = r$$

Expression: $\gcd(143, 91)$

Exact value: 13

Decimal preview: 13

Expected / target: 13

Status: **PASS**

Reasoning:

Exact rational identity

§24.1 Theta Chord

S24-1 · Repunit base R_4^(12) = $(12^4 - 1)/11 = 1885$

Expression: $(12^4 - 1) / 11$

Exact value: 1885

Decimal preview: 1885

Expected / target: 1885

Status: **PASS**

Reasoning:

Exact rational identity

$$S_{24-2} \cdot 1892 = R_4^{(12)} + p \text{ (Beethoven's Chicago } \theta)$$

Expression: $1885 + p$

Exact value: 1892

Decimal preview: 1892

Expected / target: 1892

Status: **PASS**

Reasoning:

Exact rational identity

$S_{24-3} \cdot 1893 = R_4^{(12)} + (b-1)^3$ (Tchaikovsky ς Pathétique)

Expression: $1885 + (b - 1)^3$

Exact value: 1893

Decimal preview: 1893

Expected / target: 1893

Status: **PASS**

Reasoning:

Exact rational identity

§24.1 Theta Chord — sums

$$S_{24-4} \cdot \zeta + \theta = 6 + 9 = 15 = \text{Tri}(F(5)) = p + (b-1)^3$$

Expression: $p + (b - 1)^3$

Exact value: 15

Decimal preview: 15

Expected / target: 15

Status: **PASS**

Reasoning:

Exact rational identity

§24.2 1893 alternate forms

$$S_{24-5} \cdot 1893 = b \cdot p^2 \cdot r - 2b^2 = 1911 - 18$$

Expression: $b * p^2 * \text{RadianR} - 2 * b^2$

Exact value: 1893

Decimal preview: 1893

Expected / target: 1893

Status: **PASS**

Reasoning:

Exact rational identity

$$S_{24-6} \cdot 1893 = b \cdot p \cdot \Lambda_{\text{EUV}} - (\pi + R_2)$$

Expression: $b * p * \Lambda_{\text{EUV}} - (\pi_{\text{Geom}} + R_2)$

Exact value: 1893

Decimal preview: 1893

Expected / target: 1893

Status: **PASS**

Reasoning:

Exact rational identity

$$S_{24-7} \cdot 1893 = F_5 \cdot r \cdot (R_2^2 + r) + (b-1)^3 = 5 \cdot 13 \cdot 29 + 8$$

Expression: $F(5) * \text{RadianR} * (R_2^2 + \text{RadianR}) + (b - 1)^3$

Exact value: 1893

Decimal preview: 1893

Expected / target: 1893

Status: **PASS**

Reasoning:

Exact rational identity

§24.2 — repunit factorisation

$$S_{24-8} \cdot R_4^{(12)} = 5 \cdot 13 \cdot 29 = F(5) \cdot r \cdot (R_2^2 + r) = 1885$$

Expression: $F(5) * \text{RadianR} * (R_2^2 + \text{RadianR})$

Exact value: 1885

Decimal preview: 1885

Expected / target: 1885

Status: **PASS**

Reasoning:

Exact rational identity

§24.5 Fibonacci 377

$$S_{24-9} \cdot F_{14} = 377 = 13 \cdot 29 = r \cdot (R_2^2 + r)$$

Expression: $F(14)$

Exact value: 377

Decimal preview: 377

Expected / target: 377

Status: **PASS**

Reasoning:

Exact rational identity

$S_{24-10} \cdot F_{14} = \tau + r = 364 + 13 = 377$ (calendar+radian derivation)

Expression: Tau + RadianR

Exact value: 377

Decimal preview: 377

Expected / target: 377

Status: **PASS**

Reasoning:

Exact rational identity